Supplementary Information 1: Detailed Analytical Proofs for "A Grand Unified Framework for Macroscopic Gravity-Fluid Dynamics and 0-Point Equilibrium Operation: Regularization of 3D Navier–Stokes Singularities and Proof of Global Phase Transitions via Topological Trivialization"

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Abstract

This supplementary document provides complete, self-contained mathematical derivations and step-by-step analytical proofs supporting Part 1 of the 2026 *Grand Unified Framework* series (*A Grand Unified Framework for Macroscopic Gravity-Fluid Dynamics and 0-Point Equilibrium Operation*). Specifically, we present: (i) the unit-quaternion rotational stress kinematics ($q_c \in \mathrm{Sp}(1) \cong SU(2)$), Virial boundary-to-interior tensor divergence theorem, and macroscopic L^q ($q = p + 2 \geq 3$) Korn–Sobolev enstrophy closure for the non-linear drag tensor $\boldsymbol{\mathcal{D}}_\kappa$; (ii) the Riemann–Cartan covariant formulation of the Topological Zero Tensor ($\mathcal{Z}^\mu{}_\nu \equiv 0$), Stokes–Poincaré trace normalization on the boundaryless 3-torus $\mathbb{T}^3$ ($\partial\mathbb{T}^3 = \emptyset$, $\chi(\mathbb{T}^3) = 0$), and localized zero-point core stretching cancellation ($S(\boldsymbol{u}) = 0 \implies (\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u} = \mathbf{0}$ on $K \subseteq \mathbb{T}^3$); (iii) the componentwise derivation of the localized counter-strain identity $S(\mathcal{X})\boldsymbol{\omega} = \nabla \cdot \Pi + \boldsymbol{R}$ with $\Pi_{ij} = \frac{1}{2}(\mathcal{X} \cdot \boldsymbol{\omega})\delta_{ij} + \frac{1}{2}\mathcal{X}_i\omega_j$ and $\boldsymbol{R} = -\frac{1}{2}S(\boldsymbol{\omega})\mathcal{X} - \frac{1}{4}\mathcal{X} \times (\nabla \times \boldsymbol{\omega})$ ($R_i = -\frac{1}{2}\mathcal{X}_j\partial_i\omega_j$); (iv) the unweighted advective cancellation $\langle \boldsymbol{\omega}, \nabla \cdot \Pi^{(0)} \rangle_{L^2} = 0$, transition-shell commutator bounds, and Young's inequality viscous absorption ($|\langle \boldsymbol{\omega}, \chi_\Lambda \boldsymbol{R} \rangle| \leq \frac{\nu}{2}\|\nabla\boldsymbol{\omega}\|_{L^2}^2 + \frac{M^2}{8\nu}\|\boldsymbol{\omega}\|_{L^2}^2$); and (v) the formal axiomatic equivalence between the localized core postulates on CriticalRegion(Λ), the global vorticity ceiling $|\boldsymbol{\omega}(x)| \leq \Lambda$, and the Beale–Kato–Majda (BKM) regularity criterion, synchronized 1:1 with `Supplementary_2_Formal_Verification_Lean4` and `Supplementary_3_CAS_Python_Proofs`.

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1 Macroscopic H.U.G.G.E.R Continuum Stress Closure and L^q Drag Coercivity

In this section, we provide the complete componentwise derivations for Section 2 (Equations (4)–(8)) and Section 4.2 (Equations (16)–(17)) of the main manuscript, establishing the singularity-free unit-quaternion stress kinematics, the Virial boundary-to-interior divergence bridge, and the macroscopic L^q enstrophy bound under the adaptive non-linear drag tensor $\boldsymbol{\mathcal{D}}_\kappa$.

1.1 Unit-Quaternion Rotational Stress Tensor and SI Dimensional Homogeneity

Let $\Omega \subset \mathbb{R}^3$ be a bounded macroscopic control volume with volume $V_\Omega = |\Omega| > 0$ and Lipschitz boundary $\partial\Omega$ with outward unit normal $\hat{\boldsymbol{n}}$. To represent global planetary rotation without the polar coordinate singularities ($\theta \in \{0, \pi\}$) inherent to Euler angles, let $q_c = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k} \in \mathrm{Sp}(1) \cong SU(2)$ be a unit quaternion satisfying $\|q_c\|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$, with conjugate $q_c^* = q_0 - q_1\boldsymbol{i} - q_2\boldsymbol{j} - q_3\boldsymbol{k}$.

Lemma 1.1 (Isometry and Symmetry of the Quaternion Dyadic Stress Tensor)**.** *Identifying any axial vorticity vector $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)^\mathsf{T} \in \mathbb{R}^3$ with the pure quaternion $\omega_1\boldsymbol{i} + \omega_2\boldsymbol{j} + \omega_3\boldsymbol{k} \in \mathfrak{sp}(1)$, the quaternion sandwich product $\tilde{\boldsymbol{\omega}} := q_c\boldsymbol{\omega}q_c^*$ defines an exact Euclidean rotation $\tilde{\boldsymbol{\omega}} = R(q_c)\boldsymbol{\omega}$ with $|\tilde{\boldsymbol{\omega}}|^2 = |\boldsymbol{\omega}|^2$. Consequently, for any radial density profile $\rho(r)$ ($\mathrm{kg \cdot m^{-3}}$) and angular momentum calibration factor $\mathcal{K}_\omega > 0$ ($\mathrm{m^2}$), the rotational stress tensor*

$$\boldsymbol{\sigma}_{\mathrm{rot}}(x) := \mathcal{K}_\omega \rho(r) \left(q_c\boldsymbol{\omega}q_c^*\right)^{\otimes 2}, \qquad (\boldsymbol{\sigma}_{\mathrm{rot}})_{ij} = \mathcal{K}_\omega \rho(r) \tilde{\omega}_i \tilde{\omega}_j, \tag{1}$$

is smooth across the rotational poles, symmetric ($(\boldsymbol{\sigma}_{\mathrm{rot}})_{ij} = (\boldsymbol{\sigma}_{\mathrm{rot}})_{ji}$), positive semi-definite ($\boldsymbol{\xi}^\mathsf{T}\boldsymbol{\sigma}_{\mathrm{rot}}\boldsymbol{\xi} = \mathcal{K}_\omega\rho(r)(\tilde{\boldsymbol{\omega}} \cdot \boldsymbol{\xi})^2 \geq 0$), and has trace $\mathrm{tr}(\boldsymbol{\sigma}_{\mathrm{rot}}) = \mathcal{K}_\omega\rho(r)|\boldsymbol{\omega}|^2$ with exact SI stress dimension $\mathrm{m^2 \cdot (kg \cdot m^{-3}) \cdot s^{-2} = kg \cdot m^{-1} \cdot s^{-2} = Pa}$.

Proof. Since $\mathrm{Re}(\boldsymbol{\omega}) = 0$, taking the conjugate of $\tilde{\boldsymbol{\omega}} = q_c\boldsymbol{\omega}q_c^*$ gives $\tilde{\boldsymbol{\omega}}^* = (q_c^*)^*\boldsymbol{\omega}^*q_c^* = -q_c\boldsymbol{\omega}q_c^* = -\tilde{\boldsymbol{\omega}}$, so $\tilde{\boldsymbol{\omega}}$ is again a pure quaternion (a vector in $\mathbb{R}^3$). Moreover, $|\tilde{\boldsymbol{\omega}}|^2 = \tilde{\boldsymbol{\omega}}\tilde{\boldsymbol{\omega}}^* = q_c\boldsymbol{\omega}q_c^*q_c\boldsymbol{\omega}^*q_c^* = |q_c|^2|\boldsymbol{\omega}|^2|q_c|^2 = |\boldsymbol{\omega}|^2$ because $q_cq_c^* = 1$. The dyadic square $(\tilde{\boldsymbol{\omega}} \otimes \tilde{\boldsymbol{\omega}})_{ij} = \tilde{\omega}_i\tilde{\omega}_j$ is manifestly symmetric in (i, j), and its trace is $\sum_{i=1}^3 \tilde{\omega}_i^2 = |\tilde{\boldsymbol{\omega}}|^2 = |\boldsymbol{\omega}|^2$. □

1.2 Virial Tensor Product Divergence Theorem and Macroscopic L^q Enstrophy Bound

For any continuously differentiable Cauchy stress field $\boldsymbol{\sigma} \in C^1(\bar{\Omega}; \mathbb{R}^{3\times3})$ and position vector $\boldsymbol{r} = (x_1, x_2, x_3)^\mathsf{T}$, applying the Gauss divergence theorem componentwise to the Virial boundary integral in Equations (4)–(5) of the main text yields (Appendix A of the main text):

$$\frac{1}{V_\Omega}\oint_{\partial\Omega} x_i\sigma_{jk}n_k \,\mathrm{d}\Sigma = \frac{1}{V_\Omega}\int_\Omega \partial_k(x_i\sigma_{jk}) \,\mathrm{d}V = \frac{1}{V_\Omega}\int_\Omega \left(\sigma_{ji} + x_i\partial_k\sigma_{jk}\right) \mathrm{d}V, \tag{2}$$

where $\partial_k\sigma_{jk} = (\nabla \cdot \boldsymbol{\sigma})_j$ governs the interior continuum momentum balance.

We now prove the macroscopic L^q enstrophy dissipation inequality (Equations (16)–(17) of the main text) for the incompressible v1.2 oceanic / continuum model on $\mathbb{T}^3 = [0, 2\pi)^3$.

Proposition 1.1 (Macroscopic L^q Coercivity and Enstrophy Bound under Non-Linear Drag $\boldsymbol{\mathcal{D}}_\kappa$)**.** *Consider the incompressible momentum equation on $\mathbb{T}^3$ equipped with the adaptive non-linear strain-rate viscosity $\kappa(|D\boldsymbol{u}|) = \kappa_0\left[1 + \alpha\max(0, |D\boldsymbol{u}| - s_c)^p\right]$ ($p \geq 1$, $\kappa_0 > 0$, $\alpha > 0$) from Equations (7)–(8) of the main text, where $D\boldsymbol{u} = \frac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\mathsf{T})$:*

$$\partial_t\boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} + \nabla P = \nabla \cdot \big(2\kappa(|D\boldsymbol{u}|)D\boldsymbol{u}\big), \qquad \nabla \cdot \boldsymbol{u} = 0. \tag{3}$$

Set $q := p + 2 \geq 3$. Then for any initial velocity $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ with vorticity $\boldsymbol{\omega}_0 = \nabla \times \boldsymbol{u}_0$ and every finite horizon $t > 0$:

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1. ***A Priori*** $L^q(\mathbb{T}^3 \times (0,t))$ ***Strain and Gradient Bound:***

$$\frac{1}{2}\|\boldsymbol{u}(t)\|_{L^2}^2 + 2\kappa_0 \int_0^t \|D\boldsymbol{u}(\tau)\|_{L^2}^2 \,\mathrm{d}\tau + 2\kappa_0\alpha \int_0^t \int_{\mathbb{T}^3} \max(0, |D\boldsymbol{u}| - s_c)^p |D\boldsymbol{u}|^2 \,\mathrm{d}x \,\mathrm{d}\tau = \frac{1}{2}\|\boldsymbol{u}_0\|_{L^2}^2, \quad (4)$$

which implies $\|D\boldsymbol{u}\|_{L^q(\mathbb{T}^3\times(0,t))}^q < \infty$ *and, by Korn's inequality* $\|\nabla \boldsymbol{u}\|_{L^q} \le C_K \|D\boldsymbol{u}\|_{L^q}$, $\|\nabla \boldsymbol{u}\|_{L^q(\mathbb{T}^3\times(0,t))}^q \le C_K^q \|D\boldsymbol{u}\|_{L^q(\mathbb{T}^3\times(0,t))}^q < \infty$.

2. ***Macroscopic Enstrophy Closure (Main Text Equations (16)–(17)):***

$$\|\boldsymbol{\omega}(t)\|_{L^2}^2 \le e^{2t}\left(\|\boldsymbol{\omega}_0\|_{L^2}^2 + 2C_q C_K^q \|D\boldsymbol{u}\|_{L^q(\mathbb{T}^3\times(0,t))}^q\right) < \infty. \quad (5)$$

Proof. (1) Taking the $L^2(\mathbb{T}^3)$ inner product of Equation (3) with $\boldsymbol{u}$, the convective and pressure terms vanish by $\nabla \cdot \boldsymbol{u} = 0$, and integrating $-\langle \nabla \cdot (2\kappa(|D\boldsymbol{u}|)D\boldsymbol{u}), \boldsymbol{u}\rangle$ by parts using the symmetry of $D\boldsymbol{u}$ gives $2\int_{\mathbb{T}^3} \kappa(|D\boldsymbol{u}|)|D\boldsymbol{u}|^2 \,\mathrm{d}x$. Integrating over $[0,t]$ yields Equation (4). Since $s^q = s^{p+2} \le 2^p \max(0, s - s_c)^p s^2 + (2s_c)^q$ for all $s \ge 0$, Equation (4) immediately bounds $\|D\boldsymbol{u}\|_{L^q(\mathbb{T}^3\times(0,t))}^q$ in terms of $\|\boldsymbol{u}_0\|_{L^2}^2$ and $s_c^q|\mathbb{T}^3|t$.

(2) Taking the curl of Equation (3) and testing with $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ (or testing with $-\Delta \boldsymbol{u} = \nabla \times \boldsymbol{\omega}$), the baseline molecular/eddy viscosity κ_0 contributes $2\kappa_0\|\nabla D\boldsymbol{u}\|_{L^2}^2 = \kappa_0\|\nabla \boldsymbol{\omega}\|_{L^2}^2$ (while the spatial gradient inner product of the non-linear strain flux $2\kappa_0\alpha \max(0, |D\boldsymbol{u}| - s_c)^p D\boldsymbol{u}$ is non-negative everywhere by the positive semi-definite Hessian convexity of its radially symmetric strain potential $s \mapsto \int_0^s 2\kappa_0\alpha \max(0, r - s_c)^p r \,\mathrm{d}r$, exactly analogous to Lemma 3.3 of Part 3, and may therefore be dropped from the lower bound), while the cubic vortex stretching integral satisfies $|\langle (\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u}, \boldsymbol{\omega}\rangle| \le \|\nabla \boldsymbol{u}\|_{L^3}^3$. Since $q = p + 2 \ge 3$, if $q = 3$ then $\|\nabla \boldsymbol{u}\|_{L^3}^3 = \|\nabla \boldsymbol{u}\|_{L^q}^q$, and if $q > 3$, Lebesgue interpolation between L^2 and L^q followed by Young's inequality yields $\|\nabla \boldsymbol{u}\|_{L^3}^3 \le \|\nabla \boldsymbol{u}\|_{L^2}^2 + C_q\|\nabla \boldsymbol{u}\|_{L^q}^q = \|\boldsymbol{\omega}\|_{L^2}^2 + C_q\|\nabla \boldsymbol{u}\|_{L^q}^q$. Thus:

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\boldsymbol{\omega}\|_{L^2}^2 + 2\kappa_0\|\nabla D\boldsymbol{u}\|_{L^2}^2 \le \|\boldsymbol{\omega}\|_{L^2}^2 + C_q C_K^q \|D\boldsymbol{u}\|_{L^q}^q.$$

Multiplying by 2 and applying Grönwall's differential inequality over $[0,t]$ establishes Estimate (5). □

2 Riemann–Cartan Trace Closure on $\mathbb{T}^3$ and Localized Stretching Cancellation

This section provides the detailed proofs for Section 3 (Theorem 3.1) and Section 4.1 (Equations (12)–(15)) of the main text, corresponding to `Theorem0_Basic_Topology.lean`, `Theorem1_Helmholtz_Bridge.lean`, and `CAS_Proof1_Helmholtz_Bridge.py`.

2.1 Proof of Macroscopic Trace Closure on $\mathbb{T}^3$ ($\partial\mathbb{T}^3 = \emptyset$, $\chi(\mathbb{T}^3) = 0$)

Recall from Equation (10) of the main text that the Topological Zero Tensor in mixed $(1,1)$-form on a Riemann–Cartan manifold $(\mathbb{T}^3, g, \nabla)$ is defined by

$$\mathcal{Z}^{\mu}{}_{\nu} = g^{\mu\alpha}\nabla_{(\alpha}\mathcal{X}_{\nu)} + g^{\mu\alpha}\nabla_{[\alpha}\mathcal{O}_{\nu]} - \hat{\Lambda}_0(\kappa_g)\mathcal{G}^{\mu}{}_{\nu} \equiv 0, \quad (6)$$

where $\nabla_{(\alpha}\mathcal{X}_{\nu)} = \frac{1}{2}(\nabla_\alpha \mathcal{X}_\nu + \nabla_\nu \mathcal{X}_\alpha)$, $\nabla_{[\alpha}\mathcal{O}_{\nu]} = \frac{1}{2}(\nabla_\alpha \mathcal{O}_\nu - \nabla_\nu \mathcal{O}_\alpha)$, and $\mathcal{G}_{\mu\nu} = g_{\mu\nu} + \Omega_{\mu\nu}$.

Proposition 2.1 (Three-Stage Proof of Global Trace Closure on $\mathbb{T}^3$)**.** *On the compact, oriented, boundaryless three-dimensional torus* $\mathbb{T}^3$ *(*$\partial\mathbb{T}^3 = \emptyset$*), the contracted trace* $\mathcal{Z}^{\mu}{}_{\mu} = g^{\mu\nu}\mathcal{Z}_{\mu\nu}$ *satisfies:*

$$\iiint_{\mathbb{T}^3} \mathcal{Z}^{\mu}{}_{\mu} \,\mathrm{d}V_g = \iiint_{\mathbb{T}^3} \nabla_\mu \mathcal{X}^\mu \,\mathrm{d}V_g + \iiint_{\mathbb{T}^3} g^{\mu\nu}\nabla_{[\mu}\mathcal{O}_{\nu]} \,\mathrm{d}V_g - \iiint_{\mathbb{T}^3} \hat{\Lambda}_0(\kappa_g)\mathcal{G}^{\mu}{}_{\mu} \,\mathrm{d}V_g = 0 = \chi(\mathbb{T}^3)\cdot\mathbb{E}_0. \quad (7)$$

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Proof. We evaluate the three terms consecutively:

1. **Symmetric Strain Divergence:** By metric compatibility ($\nabla_\alpha g^{\mu\alpha} = 0$) and symmetry $g^{\mu\alpha} = g^{\alpha\mu}$, $g^{\mu\alpha}\nabla_{(\alpha}\mathcal{X}_{\mu)} = \frac{1}{2}(g^{\mu\alpha}\nabla_\alpha\mathcal{X}_\mu + g^{\alpha\mu}\nabla_\mu\mathcal{X}_\alpha) = \nabla_\mu\mathcal{X}^\mu$. By Stokes' theorem on $\mathbb{T}^3$ with $\partial\mathbb{T}^3 = \emptyset$ (and independently by the solenoidal projection $\nabla_\mu\mathcal{X}^\mu = 0$ point-by-point in Equation (8) of the main text), $\iiint_{\mathbb{T}^3} \nabla_\mu\mathcal{X}^\mu \, \mathrm{d}V_g = \iint_{\partial\mathbb{T}^3} \mathcal{X}\cdot\hat{\boldsymbol{n}} \, \mathrm{d}\Sigma = 0$.

2. **Metric–Spin Orthogonality:** Because $g^{\mu\nu} = g^{\nu\mu}$ is symmetric while $\nabla_{[\mu}\mathcal{O}_{\nu]} = -\nabla_{[\nu}\mathcal{O}_{\mu]}$ is antisymmetric, relabeling dummy indices $\mu \leftrightarrow \nu$ yields $g^{\mu\nu}\nabla_{[\mu}\mathcal{O}_{\nu]} = g^{\nu\mu}\nabla_{[\nu}\mathcal{O}_{\mu]} = -g^{\mu\nu}\nabla_{[\mu}\mathcal{O}_{\nu]}$, whence $g^{\mu\nu}\nabla_{[\mu}\mathcal{O}_{\nu]} \equiv 0$ pointwise everywhere on $\mathbb{T}^3$ (mechanized as `trace_mul_sym_skw_zero` in `Theorem0_Basic_Topology.lean`).

3. **Poincaré Duality and Zero-Mean Trace Normalization:** For the closed 3-manifold $\mathbb{T}^3$, the Betti numbers are $b_0 = 1$, $b_1 = 3$, $b_2 = 3$, $b_3 = 1$, so by Poincaré duality ($b_k = b_{3-k}$), $\chi(\mathbb{T}^3) = \sum_{k=0}^{3}(-1)^k b_k(\mathbb{T}^3) = 1 - 3 + 3 - 1 = 0$. Taking the volume integral of the trace of Equation (6) combined with Steps 1 and 2 yields $\iiint_{\mathbb{T}^3} \hat{\Lambda}_0(\kappa_g)\mathcal{G}^\mu{}_\mu \, \mathrm{d}V_g = 0 = \chi(\mathbb{T}^3)\cdot\mathbb{E}_0$. □

2.2 Helmholtz Decomposition Uniqueness and Localized Stretching Cancellation on $K \subseteq \mathbb{T}^3$

Theorem 2.1 (Helmholtz Decomposition Uniqueness and Localized Core Stretching Annihilation)**.** *Let $\boldsymbol{u} \in C^2(\mathbb{T}^3;\mathbb{R}^3)$ be a divergence-free velocity field ($\nabla\cdot\boldsymbol{u} = \mathrm{tr}(\nabla\boldsymbol{u}) = 0$) with vorticity $\boldsymbol{\omega} = \nabla\times\boldsymbol{u}$. Denote the velocity gradient tensor in Cartesian coordinates by $(\nabla\boldsymbol{u})_{ij} := \partial_j u_i$, decomposed into its symmetric rate-of-strain part $S(\boldsymbol{u}) := \mathrm{sym}(\nabla\boldsymbol{u}) = \frac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\mathsf{T})$ and skew-symmetric spin part $\Omega(\boldsymbol{u}) := \mathrm{skw}(\nabla\boldsymbol{u}) = \frac{1}{2}(\nabla\boldsymbol{u} - \nabla\boldsymbol{u}^\mathsf{T})$. Then:*

1. ***Unique Symmetric–Skew Split:*** *Every 3×3 tensor $M \in \mathbb{R}^{3\times 3}$ decomposes uniquely as $M = S + A$ with $S^\mathsf{T} = S$ and $A^\mathsf{T} = -A$, namely $S = \mathrm{sym}(M)$ and $A = \mathrm{skw}(M)$.*

2. ***Universal Spin Cancellation:*** *Pointwise everywhere on $\mathbb{T}^3$, the skew-symmetric spin tensor annihilates the vorticity vector:*

$$\Omega(\boldsymbol{u})\boldsymbol{\omega} = \frac{1}{2}\boldsymbol{\omega}\times\boldsymbol{\omega} = \mathbf{0} \quad\Longrightarrow\quad (\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u} = (\nabla\boldsymbol{u})\boldsymbol{\omega} = S(\boldsymbol{u})\boldsymbol{\omega}. \tag{8}$$

3. ***Localized Zero-Point Core Stretching Cancellation (Main Text Theorem 3.1):*** *Let $K \subseteq \mathbb{T}^3$ be any localized supercritical core region on which the TZT core postulate holds: $\nabla\boldsymbol{u}(x) = \mathcal{G}_0(x)$ with $\mathrm{sym}\,\mathcal{G}_0(x) = \lambda(x)I_3$ for a scalar multiplier $\lambda : \mathbb{T}^3 \to \mathbb{R}$. Then for every $x \in K$:*

$$\lambda(x) = 0, \qquad S(\boldsymbol{u})(x) = \mathbf{0}, \qquad (\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u}(x) = S(\boldsymbol{u})(x)\boldsymbol{\omega}(x) = \mathbf{0}. \tag{9}$$

Proof. (1) If $M = S + A$ with $S^\mathsf{T} = S$ and $A^\mathsf{T} = -A$, then $M^\mathsf{T} = S - A$, so adding and subtracting gives $S = \frac{1}{2}(M + M^\mathsf{T}) = \mathrm{sym}(M)$ and $A = \frac{1}{2}(M - M^\mathsf{T}) = \mathrm{skw}(M)$ uniquely (`helmholtz_unique` in `Theorem1_Helmholtz_Bridge.lean`).

(2) With $(\nabla\boldsymbol{u})_{ij} = \partial_j u_i$ and $\boldsymbol{\omega} = (\partial_2 u_3 - \partial_3 u_2,\, \partial_3 u_1 - \partial_1 u_3,\, \partial_1 u_2 - \partial_2 u_1)^\mathsf{T}$, the skew matrix is

$$\Omega(\boldsymbol{u}) = \frac{1}{2}\begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}.$$

Multiplying $\Omega(\boldsymbol{u})$ by $\boldsymbol{\omega} = (\omega_1,\omega_2,\omega_3)^\mathsf{T}$ yields the first component $\frac{1}{2}(-\omega_3\omega_2 + \omega_2\omega_3) = 0$, the second component $\frac{1}{2}(\omega_3\omega_1 - \omega_1\omega_3) = 0$, and the third component $\frac{1}{2}(-\omega_2\omega_1 + \omega_1\omega_2) = 0$. Since $((\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u})_i = \omega_j\partial_j u_i = (\nabla\boldsymbol{u})_{ij}\omega_j = (S(\boldsymbol{u})\boldsymbol{\omega} + \Omega(\boldsymbol{u})\boldsymbol{\omega})_i$, Equation (8) holds identically (`H1_of_reconstruction` in `Theorem1_Helmholtz_Bridge.lean` and `CAS_Proof1_Helmholtz_Bridge.py`).

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(3) For any $x \in K$, combining $\nabla \boldsymbol{u}(x) = \mathcal{G}_0(x)$ with $\operatorname{sym} \mathcal{G}_0(x) = \lambda(x) I_3$ gives $S(\boldsymbol{u})(x) = \operatorname{sym}(\nabla \boldsymbol{u}(x)) = \lambda(x) I_3$. Taking the matrix trace yields $\operatorname{tr}(S(\boldsymbol{u})(x)) = \operatorname{tr}(\lambda(x) I_3) = 3\lambda(x)$. On the other hand, by incompressibility, $\operatorname{tr}(S(\boldsymbol{u})(x)) = \operatorname{tr}(\nabla \boldsymbol{u}(x)) = \nabla \cdot \boldsymbol{u}(x) = 0$. Therefore $3\lambda(x) = 0 \implies \lambda(x) = 0$, which forces $S(\boldsymbol{u})(x) = \mathbf{0}$ and $(\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u}(x) = S(\boldsymbol{u})(x)\boldsymbol{\omega}(x) = \mathbf{0}$ for all $x \in K$ (`lambda_from_incompressibility`, `strain_vanishes_on_core`, and `stretching_vanishes_on_core` in `Theorem1_Helmholtz_Bridge.lean` and `CAS_Proof1_Helmholtz_Bridge.py`). □

3 Componentwise Derivation of the Counter-Strain Identity $S(\mathcal{X})\boldsymbol{\omega} = \nabla \cdot \Pi + \boldsymbol{R}$

This section provides the step-by-step tensor-index derivation of Equation (21) in Section 4.2 of the main text, corresponding to `Theorem2_Integral_Alignment.lean`, `CAS_Proof2_Integral_Alignment.py`, and `CAS_Proof4_Viscous_Absorption.py`.

3.1 Localized TZT Momentum Forcing and Convex Enstrophy Partition

Recall from Equations (18)–(19) of the main text that the TZT-regularized incompressible momentum equation on $\mathbb{T}^3$ is

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nu \Delta \boldsymbol{u} + \mathcal{F}_{\mathrm{TZT}}, \qquad \nabla \cdot \boldsymbol{u} = 0, \tag{10}$$

where $\chi_\Lambda(|\boldsymbol{\omega}|^2) \in C^\infty(\mathbb{R}_{\geq 0}; [0,1])$ is a smooth cutoff satisfying $\chi_\Lambda(s) = 0$ for $s \leq \Lambda^2$ and $\chi_\Lambda(s) = 1$ for $s \geq (\Lambda + \delta)^2$, $\mathcal{B} := \nabla \times (-\Delta)^{-1}$ is the periodic Biot–Savart inverse-curl operator on $\mathbb{T}^3$ (satisfying $\nabla \cdot \mathcal{B}(\boldsymbol{J}) = 0$ and $\nabla \times \mathcal{B}(\boldsymbol{J}) = \mathbb{P}\boldsymbol{J}$), and

$$\begin{aligned} \mathcal{F}_{\mathrm{TZT}} &:= \mathcal{B}\big[\chi_\Lambda(|\boldsymbol{\omega}|^2)\big(S(\mathcal{X})\boldsymbol{\omega} - S(\boldsymbol{u})\boldsymbol{\omega}\big)\big] - \nabla \times \big(\chi_\Lambda(|\boldsymbol{\omega}|^2)\, \boldsymbol{\mathcal{O}}^{\mathrm{turb}} \cdot \boldsymbol{\omega}\big), \\ \mathcal{X}_i &= (\mathbb{P}_\ell)_i^{\ j}\left(-C_X \frac{\partial_j |\boldsymbol{\omega}|^2}{|\nabla |\boldsymbol{\omega}|^2| + \delta_m}\right), \qquad \mathcal{O}_{ij}^{\mathrm{turb}} = C_O \nu_{\mathrm{turb}} \frac{\omega_i \omega_j}{|\boldsymbol{\omega}|^2 + \varepsilon}. \end{aligned} \tag{11}$$

Because the bounded vector field $\boldsymbol{g} = -C_X \nabla |\boldsymbol{\omega}|^2 / (|\nabla |\boldsymbol{\omega}|^2| + \delta_m)$ ($|\boldsymbol{g}(x)| \leq C_X$) is convolved with the smooth mollifier J_ℓ at fixed scale $\ell > 0$ inside $\mathbb{P}_\ell = \mathbb{P} J_\ell$, $\mathcal{X}$ is strictly solenoidal ($\nabla \cdot \mathcal{X} = 0$) and satisfies the uniform L^∞ bound $\|\mathcal{X}\|_{L^\infty(\mathbb{T}^3)} \leq M := C_\ell C_X < \infty$.

Taking the curl of Equation (10) yields the vorticity transport equation

$$\partial_t \boldsymbol{\omega} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{\omega} = S(\boldsymbol{u})\boldsymbol{\omega} + \nu \Delta \boldsymbol{\omega} + \nabla \times \mathcal{F}_{\mathrm{TZT}},$$

where $\nabla \times \mathcal{F}_{\mathrm{TZT}} = \mathbb{P}\big[\chi_\Lambda\big(S(\mathcal{X})\boldsymbol{\omega} - S(\boldsymbol{u})\boldsymbol{\omega}\big)\big] - \nabla \times \nabla \times (\chi_\Lambda \boldsymbol{\mathcal{O}}^{\mathrm{turb}} \cdot \boldsymbol{\omega})$. Taking the $L^2(\mathbb{T}^3)$ inner product with $\boldsymbol{\omega}$ and using $\langle \boldsymbol{\omega}, (\boldsymbol{u} \cdot \nabla)\boldsymbol{\omega} \rangle = 0$ along with the self-adjointness of $\mathbb{P}$ and $\mathbb{P}\boldsymbol{\omega} = \boldsymbol{\omega}$ (since $\nabla \cdot \boldsymbol{\omega} = 0$) produces the exact convex enstrophy production split (Equation (20) of the main text):

$$\begin{aligned} \langle \boldsymbol{\omega}, S(\boldsymbol{u})\boldsymbol{\omega} + \nabla \times \mathcal{F}_{\mathrm{TZT}} \rangle = &\int_{\mathbb{T}^3} \big(1 - \chi_\Lambda(|\boldsymbol{\omega}|^2)\big)\, \boldsymbol{\omega} \cdot S(\boldsymbol{u})\boldsymbol{\omega}\, \mathrm{d}x + \int_{\mathbb{T}^3} \chi_\Lambda(|\boldsymbol{\omega}|^2)\, \boldsymbol{\omega} \cdot S(\mathcal{X})\boldsymbol{\omega}\, \mathrm{d}x \\ &- \int_{\mathbb{T}^3} (\nabla \times \boldsymbol{\omega}) \cdot \nabla \times \big(\chi_\Lambda(|\boldsymbol{\omega}|^2)\, \boldsymbol{\mathcal{O}}^{\mathrm{turb}} \cdot \boldsymbol{\omega}\big)\, \mathrm{d}x. \end{aligned} \tag{12}$$

On the subcritical support $\operatorname{supp}(1 - \chi_\Lambda)$ where $|\boldsymbol{\omega}(x)| \leq \Lambda + \delta$, Cauchy–Schwarz and the periodic Calderón–Zygmund identity $\|\nabla \boldsymbol{u}\|_{L^2(\mathbb{T}^3)} = \|\boldsymbol{\omega}\|_{L^2(\mathbb{T}^3)}$ immediately give:

$$\left| \int_{\mathbb{T}^3} \big(1 - \chi_\Lambda(|\boldsymbol{\omega}|^2)\big)\, \boldsymbol{\omega} \cdot S(\boldsymbol{u})\boldsymbol{\omega}\, \mathrm{d}x \right| \leq (\Lambda + \delta) \|\boldsymbol{\omega}\|_{L^2} \|S(\boldsymbol{u})\|_{L^2} \leq (\Lambda + \delta) \|\boldsymbol{\omega}\|_{L^2}^2. \tag{13}$$

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3.2 Pointwise Index Proof of the Counter-Strain Identity (Main Text Equation (21))

We now prove the exact pointwise decomposition of $S(\mathcal{X})\boldsymbol{\omega}$ on the supercritical support.

Theorem 3.1 (Exact Pointwise Counter-Strain Divergence–Remainder Identity)**.** *Let $\mathcal{X}, \boldsymbol{\omega} \in C^1(\mathbb{T}^3;\mathbb{R}^3)$ be divergence-free vector fields ($\partial_j \mathcal{X}_j = 0$ and $\partial_j \omega_j = 0$). Adopting the standard tensor divergence convention $(\nabla \cdot \Pi)_i := \partial_j \Pi_{ij}$, define the rank-2 flux tensor Π and the remainder vector $\boldsymbol{R}$ by*

$$\Pi_{ij} := \frac{1}{2}(\mathcal{X} \cdot \boldsymbol{\omega})\delta_{ij} + \frac{1}{2}\mathcal{X}_i \omega_j, \qquad \boldsymbol{R} := -\frac{1}{2}S(\boldsymbol{\omega})\mathcal{X} - \frac{1}{4}\mathcal{X} \times (\nabla \times \boldsymbol{\omega}). \tag{14}$$

Then the following identities hold pointwise everywhere on $\mathbb{T}^3$:

1. ***Compact directional derivative form of R:*** *For each component $i \in \{1,2,3\}$,*

$$R_i = -\frac{1}{2}\mathcal{X}_j \partial_i \omega_j = -\frac{1}{2}\big((\nabla \boldsymbol{\omega})^{\mathsf{T}} \mathcal{X}\big)_i. \tag{15}$$

2. ***Exact Counter-Strain Split:***

$$S(\mathcal{X})\boldsymbol{\omega} = \nabla \cdot \Pi + \boldsymbol{R}. \tag{16}$$

3. ***Vanishing of R on the Idealized Zero-Point Core Interior*** $\mathrm{int}(K)$***:*** *Whenever $\boldsymbol{u} \in C^3(\mathbb{T}^3;\mathbb{R}^3)$ satisfies the zero-point core postulate $S(\boldsymbol{u}) = \mathbf{0}$ on a subset $K \subseteq \mathbb{T}^3$, every interior point $x \in \mathrm{int}(K)$ satisfies $\nabla \boldsymbol{\omega}(x) = \mathbf{0}$, $\nabla \times \boldsymbol{\omega}(x) = \mathbf{0}$, and $\boldsymbol{R}(x) = \mathbf{0}$.*

Proof. (1) By definition of the symmetric vorticity gradient $S(\boldsymbol{\omega})_{ij} = \frac{1}{2}(\partial_i \omega_j + \partial_j \omega_i)$, the i-th component of $-\frac{1}{2}S(\boldsymbol{\omega})\mathcal{X}$ is

$$-\frac{1}{2}\big(S(\boldsymbol{\omega})\mathcal{X}\big)_i = -\frac{1}{4}(\partial_i \omega_j + \partial_j \omega_i)\mathcal{X}_j.$$

Using the Levi-Civita contraction identity $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$, the i-th component of the cross product $\mathcal{X} \times (\nabla \times \boldsymbol{\omega})$ is

$$\big(\mathcal{X} \times (\nabla \times \boldsymbol{\omega})\big)_i = \varepsilon_{ijk}\mathcal{X}_j \varepsilon_{klm}\partial_l \omega_m = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\mathcal{X}_j \partial_l \omega_m = \mathcal{X}_j(\partial_i \omega_j - \partial_j \omega_i).$$

Multiplying by $-\frac{1}{4}$ and adding to $-\frac{1}{2}(S(\boldsymbol{\omega})\mathcal{X})_i$ cancels the $\partial_j \omega_i$ terms identically:

$$R_i = -\frac{1}{4}(\partial_i \omega_j + \partial_j \omega_i)\mathcal{X}_j - \frac{1}{4}\mathcal{X}_j(\partial_i \omega_j - \partial_j \omega_i) = -\frac{1}{2}\mathcal{X}_j \partial_i \omega_j,$$

which proves Equation (15) (verified symbolically in `CAS_Proof2_Integral_Alignment.py` and `CAS_Proof4_Viscous_Absorption.py`).

(2) Expanding the divergence $(\nabla \cdot \Pi)_i = \partial_j \Pi_{ij}$ using the product rule gives:

$$\begin{aligned}(\nabla \cdot \Pi)_i &= \frac{1}{2}\partial_j\big((\mathcal{X}_k \omega_k)\delta_{ij}\big) + \frac{1}{2}\partial_j(\mathcal{X}_i \omega_j) \\ &= \frac{1}{2}\partial_i(\mathcal{X}_j \omega_j) + \frac{1}{2}(\partial_j \mathcal{X}_i)\omega_j + \frac{1}{2}\mathcal{X}_i(\partial_j \omega_j) \\ &= \frac{1}{2}(\partial_i \mathcal{X}_j)\omega_j + \frac{1}{2}\mathcal{X}_j(\partial_i \omega_j) + \frac{1}{2}(\partial_j \mathcal{X}_i)\omega_j + \frac{1}{2}\mathcal{X}_i(\nabla \cdot \boldsymbol{\omega}).\end{aligned}$$

Since $\nabla \cdot \boldsymbol{\omega} = \partial_j \omega_j = 0$, the last term vanishes. Adding $R_i = -\frac{1}{2}\mathcal{X}_j(\partial_i \omega_j)$ cancels the $+\frac{1}{2}\mathcal{X}_j(\partial_i \omega_j)$ term, leaving:

$$(\nabla \cdot \Pi)_i + R_i = \frac{1}{2}(\partial_i \mathcal{X}_j + \partial_j \mathcal{X}_i)\omega_j = \big(S(\mathcal{X})\boldsymbol{\omega}\big)_i,$$

which establishes Equation (16).

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(3) For any C^2 vector field $\boldsymbol{u}$, the spatial derivatives of vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ are linear combinations of the first partial derivatives of the symmetric strain $S(\boldsymbol{u})$ via the classic Saint-Venant / Korn compatibility identity:

$$\partial_k \omega_i = \varepsilon_{ilm} \partial_k \partial_l u_m = \varepsilon_{ilm} \big(\partial_l S_{km}(\boldsymbol{u}) - \partial_m S_{kl}(\boldsymbol{u}) \big).$$

On the open interior $\mathrm{int}(K)$ where $S(\boldsymbol{u}) \equiv \mathbf{0}$, all partial derivatives $\partial_l S_{km}(\boldsymbol{u})(x) = 0$ vanish, so $\nabla \boldsymbol{\omega}(x) = \mathbf{0}$, $\nabla \times \boldsymbol{\omega}(x) = \mathbf{0}$, and $\boldsymbol{R}(x) = \mathbf{0}$ (`hessian_eq_zero_of_eventually_strain_zero`, `remainder_vanishes_on_core_interior`, and `H2_on_core_interior` in `Theorem2_Integral_Alignment.lean`). □

4 Advective Flux Cancellation, Killing Rigidity, and Young's Viscous Lock

This section provides the complete proofs for the flux integral estimates and Equation (22) in Section 4.2 of the main text, corresponding to `Theorem3_Mean_Conserved.lean`, `Theorem4_Viscous_Absorption.lean`, `CAS_Proof3_Mean_Conserved.py`, and `CAS_Proof4_Viscous_Absorption.py`.

4.1 Unweighted Advective Cancellation $\langle \boldsymbol{\omega}, \nabla \cdot \Pi^{(0)} \rangle = 0$ and Shell Commutators

Proposition 4.1 (Flux Decomposition and Integration by Parts on $\mathbb{T}^3$)**.** *Decompose the tensor $\Pi_{ij} = \frac{1}{2}(\mathcal{X} \cdot \boldsymbol{\omega})\delta_{ij} + \frac{1}{2}\mathcal{X}_i \omega_j$ into its advective-pressure part $\Pi^{(0)}_{ij}$ and skew-exchange part Π^{skw}_{ij}:*

$$\Pi_{ij} = \Pi^{(0)}_{ij} + \Pi^{\mathrm{skw}}_{ij}, \qquad \Pi^{(0)}_{ij} := \frac{1}{2}(\mathcal{X} \cdot \boldsymbol{\omega})\delta_{ij} + \frac{1}{2}\omega_i \mathcal{X}_j, \qquad \Pi^{\mathrm{skw}}_{ij} := \frac{1}{2}(\mathcal{X}_i \omega_j - \omega_i \mathcal{X}_j). \tag{17}$$

Then for any solenoidal fields $\mathcal{X}, \boldsymbol{\omega} \in H^1_\sigma(\mathbb{T}^3)$ ($\nabla \cdot \mathcal{X} = 0$, $\nabla \cdot \boldsymbol{\omega} = 0$) with $\|\mathcal{X}\|_{L^\infty(\mathbb{T}^3)} \le M$:

1. ***Pointwise Advective Divergence Identity:***

$$\omega_i \partial_j \Pi^{(0)}_{ij} = \frac{1}{2}\partial_i\big(\omega_i(\mathcal{X} \cdot \boldsymbol{\omega})\big) + \frac{1}{4}\partial_j\big(\mathcal{X}_j |\boldsymbol{\omega}|^2\big). \tag{18}$$

   Consequently, on the periodic boundaryless torus $\mathbb{T}^3$, $\langle \boldsymbol{\omega}, \nabla \cdot \Pi^{(0)} \rangle_{L^2(\mathbb{T}^3)} = \int_{\mathbb{T}^3} \omega_i \partial_j \Pi^{(0)}_{ij} \, \mathrm{d}x = 0$.

2. ***Weighted Cutoff Commutator and Skew-Flux Bounds:*** *With smooth cutoff $\chi_\Lambda = \chi_\Lambda(|\boldsymbol{\omega}|^2)$,*

$$\begin{aligned} \left| \int_{\mathbb{T}^3} \chi_\Lambda \omega_i \partial_j \Pi^{(0)}_{ij} \, \mathrm{d}x \right| &= \left| -\frac{1}{2}\int_{\mathbb{T}^3} (\boldsymbol{\omega} \cdot \nabla \chi_\Lambda)(\mathcal{X} \cdot \boldsymbol{\omega}) \, \mathrm{d}x - \frac{1}{4}\int_{\mathbb{T}^3} (\mathcal{X} \cdot \nabla \chi_\Lambda)|\boldsymbol{\omega}|^2 \, \mathrm{d}x \right| \\ &\le C_{\Lambda,\delta} M \|\boldsymbol{\omega}\|_{L^2(\mathbb{T}^3)} \|\nabla \boldsymbol{\omega}\|_{L^2(\mathbb{T}^3)}, \end{aligned} \tag{19}$$

$$\left| \int_{\mathbb{T}^3} \chi_\Lambda \omega_i \partial_j \Pi^{\mathrm{skw}}_{ij} \, \mathrm{d}x \right| = \left| -\frac{1}{2}\int_{\mathbb{T}^3} \chi_\Lambda (\partial_j \omega_i) \mathcal{X}_i \omega_j \, \mathrm{d}x \right| \le \frac{M}{2} \|\boldsymbol{\omega}\|_{L^2(\mathbb{T}^3)} \|\nabla \boldsymbol{\omega}\|_{L^2(\mathbb{T}^3)}, \tag{20}$$

   where $C_{\Lambda,\delta} := \frac{3}{2}\|\chi'_\Lambda\|_{L^\infty}(\Lambda + \delta)^2 < \infty$ because $\mathrm{supp}(\nabla \chi_\Lambda)$ is strictly contained in the bounded transition shell $\Lambda \le |\boldsymbol{\omega}(x)| \le \Lambda + \delta$.

Proof. (1) Since $\partial_j \mathcal{X}_j = 0$ and $\partial_i \omega_i = 0$, expanding the right-hand side of Equation (18) yields

$$\frac{1}{2}(\partial_i \omega_i)(\mathcal{X} \cdot \boldsymbol{\omega}) + \frac{1}{2}\omega_i \partial_i (\mathcal{X} \cdot \boldsymbol{\omega}) + \frac{1}{4}(\partial_j \mathcal{X}_j)|\boldsymbol{\omega}|^2 + \frac{1}{2}\mathcal{X}_j \omega_i \partial_j \omega_i = \omega_i \left[\frac{1}{2}\partial_i (\mathcal{X} \cdot \boldsymbol{\omega}) + \frac{1}{2}\mathcal{X}_j \partial_j \omega_i \right] = \omega_i \partial_j \Pi^{(0)}_{ij}.$$

Integrating the exact divergence (18) over $\mathbb{T}^3$ ($\partial \mathbb{T}^3 = \emptyset$) gives 0 identically (verified symbolically in `CAS_Proof4_Viscous_Absorption.py`).

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(2) Multiplying Equation (18) by $\chi_\Lambda(|\boldsymbol{\omega}|^2)$ and integrating by parts transfers the derivative onto $\nabla\chi_\Lambda = 2\chi'_\Lambda(|\boldsymbol{\omega}|^2)(\nabla\boldsymbol{\omega})^{\mathsf{T}}\boldsymbol{\omega}$. Because $\chi'_\Lambda(|\boldsymbol{\omega}|^2) = 0$ whenever $|\boldsymbol{\omega}| < \Lambda$ or $|\boldsymbol{\omega}| > \Lambda + \delta$, on $\operatorname{supp}(\nabla\chi_\Lambda)$ we have $|\boldsymbol{\omega}(x)| \le \Lambda + \delta$ and $|\nabla\chi_\Lambda(x)| \le 2\|\chi'_\Lambda\|_{L^\infty}(\Lambda + \delta)|\nabla\boldsymbol{\omega}(x)|$. Applying Hölder's inequality yields Estimate (19). For $\Pi^{\mathrm{skw}}_{ij} = \frac{1}{2}(\mathcal{X}_i\omega_j - \omega_i\mathcal{X}_j)$, integrating by parts gives

$$\int_{\mathbb{T}^3} \chi_\Lambda \omega_i \partial_j \Pi^{\mathrm{skw}}_{ij}\,\mathrm{d}x = -\frac{1}{2}\int_{\mathbb{T}^3} (\omega_i\partial_j\chi_\Lambda + \chi_\Lambda\partial_j\omega_i)(\mathcal{X}_i\omega_j - \omega_i\mathcal{X}_j)\,\mathrm{d}x.$$

In the first term, $\omega_i(\mathcal{X}_i\omega_j - \omega_i\mathcal{X}_j) = (\boldsymbol{\omega}\cdot\mathcal{X})\omega_j - |\boldsymbol{\omega}|^2\mathcal{X}_j$ is supported on the bounded shell $\Lambda \le |\boldsymbol{\omega}| \le \Lambda + \delta$, while on the core interior where $\partial_j\chi_\Lambda = 0$, $(\partial_j\omega_i)\omega_i\mathcal{X}_j = \frac{1}{2}\mathcal{X}_j\partial_j|\boldsymbol{\omega}|^2 = \frac{1}{2}\mathcal{X}\cdot\nabla|\boldsymbol{\omega}|^2$, leaving $-\frac{1}{2}\int_{\mathbb{T}^3}\chi_\Lambda(\partial_j\omega_i)\mathcal{X}_i\omega_j\,\mathrm{d}x$ bounded by $\frac{M}{2}\|\boldsymbol{\omega}\|_{L^2}\|\nabla\boldsymbol{\omega}\|_{L^2}$. □

4.2 Killing Rigidity on $\mathbb{T}^3$ and Young's Inequality Viscous Lock

Proposition 4.2 (Killing Rigidity and Avoidance of Global Over-Constraint)**.** *Let $\mathcal{X} \in C^1(\mathbb{T}^3;\mathbb{R}^3)$. If both the symmetric strain $S(\mathcal{X}) = \operatorname{sym}(\nabla\mathcal{X}) = \mathbf{0}$ and the skew spin $\Omega(\mathcal{X}) = \operatorname{skw}(\nabla\mathcal{X}) = \mathbf{0}$ vanish globally on $\mathbb{T}^3$ (or by periodic Korn's inequality $\|\nabla\mathcal{X}\|^2_{L^2} = 2\|S(\mathcal{X})\|^2_{L^2}$, if $S(\mathcal{X}) = \mathbf{0}$ globally on $\mathbb{T}^3$), then $\nabla\mathcal{X} \equiv \mathbf{0}$ and $\mathcal{X}(x) \equiv \mathcal{X}(0)$ is spatially constant.*

Proof. Pointwise, $\nabla\mathcal{X} = S(\mathcal{X}) + \Omega(\mathcal{X}) = \mathbf{0} + \mathbf{0} = \mathbf{0}$. Since $\mathbb{T}^3$ is path-connected, the mean value theorem implies $\mathcal{X}(x) = \mathcal{X}(0)$ for all $x \in \mathbb{T}^3$ (`const_of_grad_eq_zero`, `X_constant_of_killing`, and `X_constant_of_global_core` in `Theorem3_Mean_Conserved.lean`). This rigorously justifies why the TZT counter-strain projection is localized to the supercritical subset $K = \mathrm{CriticalRegion}(\Lambda) \subsetneq \mathbb{T}^3$ via $\chi_\Lambda(|\boldsymbol{\omega}|^2)$ rather than imposed globally across all of $\mathbb{T}^3$. □

Theorem 4.1 (Young's Inequality Viscous Absorption and Grönwall Enstrophy Bound)**.** *For any kinematic viscosity $\nu > 0$ ($\nu \ne 0$) and any non-trivial intake bound $M > 0$ ($M \ne 0$) with $\|\mathcal{X}\|_{L^\infty(\mathbb{T}^3)} \le M$, the remainder inner product $\langle\boldsymbol{\omega}, \chi_\Lambda\boldsymbol{R}\rangle$ (where $R_i = -\frac{1}{2}\mathcal{X}_j\partial_i\omega_j$) satisfies Main Text Equation (22):*

$$\left|\langle\boldsymbol{\omega}, \chi_\Lambda\boldsymbol{R}\rangle\right| = \left|-\frac{1}{2}\int_{\mathbb{T}^3}\chi_\Lambda\omega_i\mathcal{X}_j\partial_i\omega_j\,\mathrm{d}x\right| \le \frac{M}{2}\|\boldsymbol{\omega}\|_{L^2}\|\nabla\boldsymbol{\omega}\|_{L^2} \le \frac{\nu}{2}\|\nabla\boldsymbol{\omega}\|^2_{L^2} + \frac{M^2}{8\nu}\|\boldsymbol{\omega}\|^2_{L^2}. \tag{21}$$

More generally, absorbing the combined remainder, skew-flux, and transition-shell commutator bound $C_{\mathrm{tot}}\|\boldsymbol{\omega}\|_{L^2}\|\nabla\boldsymbol{\omega}\|_{L^2}$ into half the molecular enstrophy dissipation $\frac{\nu}{2}\|\nabla\boldsymbol{\omega}\|^2_{L^2}$ leaves a strictly linear enstrophy growth rate $\gamma_{\mathrm{TZT}} := 2(\Lambda + \delta) + \frac{C^2_{\mathrm{tot}}}{\nu} < \infty$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\boldsymbol{\omega}(t)\|^2_{L^2} + \nu\|\nabla\boldsymbol{\omega}(t)\|^2_{L^2} \le \gamma_{\mathrm{TZT}}\|\boldsymbol{\omega}(t)\|^2_{L^2} \quad\Longrightarrow\quad \|\boldsymbol{\omega}(t)\|^2_{L^2} \le \|\boldsymbol{\omega}(0)\|^2_{L^2}\,e^{\gamma_{\mathrm{TZT}}t} < \infty. \tag{22}$$

Proof. Since $0 \le \chi_\Lambda \le 1$ and $|\mathcal{X}(x)| \le M$, the Cauchy–Schwarz inequality gives

$$\left|-\frac{1}{2}\int_{\mathbb{T}^3}\chi_\Lambda\omega_i\mathcal{X}_j\partial_i\omega_j\,\mathrm{d}x\right| \le \frac{1}{2}\int_{\mathbb{T}^3}|\boldsymbol{\omega}(x)||\mathcal{X}(x)||\nabla\boldsymbol{\omega}(x)|\,\mathrm{d}x \le \frac{M}{2}\|\boldsymbol{\omega}\|_{L^2}\|\nabla\boldsymbol{\omega}\|_{L^2}.$$

For any real numbers $A = \|\nabla\boldsymbol{\omega}\|_{L^2}$, $B = \|\boldsymbol{\omega}\|_{L^2}$, and $\nu > 0$, completing the square yields the exact algebraic identity:

$$\frac{\nu}{2}A^2 + \frac{M^2}{8\nu}B^2 - \frac{M}{2}AB = \frac{1}{2\nu}\left(\nu A - \frac{M}{2}B\right)^2 \ge 0,$$

which proves Inequality (21) for the full remainder vector $R_i = -\frac{1}{2}\mathcal{X}_j\partial_i\omega_j$ with prefactor $\frac{M}{2}$. (Note: When restricted strictly to the curl channel $\boldsymbol{R}_{\mathrm{skew}} = -\frac{1}{4}\mathcal{X}\times(\nabla\times\boldsymbol{\omega})$ under the alignment condition $S(\boldsymbol{\omega})\mathcal{X} = \mathbf{0}$, as parameterized in `young_enstrophy` and `viscous_absorption` of `Theorem4_Viscous_Absorption.lean` and Section 2.4 of `CAS_Proof4_Viscous_Absorption.py`, replacing the prefactor $\frac{M}{2}$ by $\frac{M}{4}$ with $\varepsilon^* = \frac{M^2}{32\nu}$ yields $\frac{\nu}{2}A^2 + \varepsilon^*B^2 - \frac{M}{4}AB = \frac{1}{\varepsilon^*}\left(\varepsilon^*B - \frac{M}{8}A\right)^2 \ge 0$ and differential rate $K_\nu = 2\varepsilon^* = \frac{M^2}{16\nu}$.) Subtracting $\frac{\nu}{2}\|\nabla\boldsymbol{\omega}\|^2_{L^2}$ from the molecular viscous dissipation $\nu\|\nabla\boldsymbol{\omega}\|^2_{L^2}$, multiplying by 2, and integrating $\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-\gamma_{\mathrm{TZT}}t}\|\boldsymbol{\omega}(t)\|^2_{L^2}\right) \le 0$ (`gronwall_scalar` and `enstrophy_gronwall_L2` in `Theorem4_Viscous_Absorption.lean`) yields Estimate (22). □

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5 Axiomatic Core Equivalence, BKM Regularity Bound, and Non-Vacuity Witness

In this section, we prove the exact logical equivalence between the localized zero-point core postulates (`TZTPostulates s lam K`) and the global pointwise vorticity ceiling $|\boldsymbol{\omega}(x)| \leq \Lambda$ mechanized in Section 5.2 of the main text (`Theorem5_Consistency_Audit.lean` and `Theorem6_Vorticity_Ceiling.lean`).

Definition 5.1 (Localized Zero-Point Core Postulate System on $K \subseteq \mathbb{T}^3$)**.** For a velocity state $s = (\boldsymbol{u}, \boldsymbol{\omega}, \mathcal{G}_0, \mathcal{X})$ on $\mathbb{T}^3$, a scalar multiplier $\lambda : \mathbb{T}^3 \to \mathbb{R}$, and a localized subset $K \subseteq \mathbb{T}^3$, the predicate $\mathrm{TZTPostulates}(s, \lambda, K)$ holds if and only if:

1. $\boldsymbol{u} \in C^2(\mathbb{T}^3; \mathbb{R}^3)$ is 2π-periodic and divergence-free ($\nabla \cdot \boldsymbol{u}(x) = 0$ for all $x \in \mathbb{T}^3$), with $\boldsymbol{\omega}(x) = \nabla \times \boldsymbol{u}(x)$;
2. For every $x \in K$, $\nabla \boldsymbol{u}(x) = \mathcal{G}_0(x)$, $\operatorname{sym} \mathcal{G}_0(x) = \lambda(x) I_3$, and $\operatorname{skw} \mathcal{G}_0(x) = \mathbf{0}$.

For any threshold $\Lambda \geq 0$, the supercritical set is defined by $\mathrm{CriticalRegion}(s, \Lambda) := \{x \in \mathbb{T}^3 : |\boldsymbol{\omega}(x)|^2 > \Lambda^2\}$.

Theorem 5.1 (Logical Equivalence to Global Vorticity Ceiling, BKM Bound, and Kolmogorov Witness)**.** *Let $\boldsymbol{u} \in C^2(\mathbb{T}^3; \mathbb{R}^3)$ be periodic and divergence-free ($\nabla \cdot \boldsymbol{u} = 0$), and let $\Lambda \geq 0$. Then:*

1. ***Core Vorticity Annihilation:*** *If $\mathrm{TZTPostulates}(s, \lambda, K)$ holds, then for every $x \in K$, $\nabla \boldsymbol{u}(x) = \mathbf{0}$ and $\boldsymbol{\omega}(x) = \mathbf{0}$.*
2. ***Vorticity Ceiling Equivalence (`vorticity_ceiling` & `ceiling_iff` in `Theorem6`):***
$$\Big(\exists\, \lambda : \mathbb{T}^3 \to \mathbb{R},\ \mathrm{TZTPostulates}\big(s_{\boldsymbol{u}}, \lambda, \mathrm{CriticalRegion}(s_{\boldsymbol{u}}, \Lambda)\big)\Big) \quad \Longleftrightarrow \quad \forall\, x \in \mathbb{T}^3,\ |\boldsymbol{\omega}(x)|^2 \leq \Lambda^2. \tag{23}$$
3. ***Beale–Kato–Majda (BKM) Integral Bound (`BKM_sup_bound` in `Theorem6`):*** *If $\sup_{x \in \mathbb{T}^3} |\boldsymbol{\omega}(x,t)| \leq \Lambda$ for all $t \in [0, T]$ ($T \geq 0$), then*
$$\int_0^T \|\boldsymbol{\omega}(t)\|_{L^\infty(\mathbb{T}^3)}\, \mathrm{d}t \leq T \sup_{t \in [0,T]} \|\boldsymbol{\omega}(t)\|_{L^\infty(\mathbb{T}^3)} \leq \Lambda T < \infty. \tag{24}$$
4. ***Non-Trivial Satisfiability Witness (`localized_postulates_admit_vorticity` in `Theorem5`):*** *The smooth periodic Kolmogorov shear flow $\boldsymbol{u}^*(x) := (\sin x_2, 0, 0)^{\mathsf{T}}$ satisfies $\nabla \cdot \boldsymbol{u}^* = 0$, has non-vanishing vorticity $\boldsymbol{\omega}^*(x) = (0, 0, -\cos x_2)^{\mathsf{T}}$ ($|\boldsymbol{\omega}^*(0)|^2 = 1 > 0$), and satisfies $\mathrm{TZTPostulates}(s_{\boldsymbol{u}^*}, 0, \mathrm{CriticalRegion}(s_{\boldsymbol{u}^*}, \Lambda))$ for every $\Lambda \geq 1$.*

Proof. (1) For any $x \in K$, Theorem 2.1(3) gives $\lambda(x) = 0$ and $\operatorname{sym}(\nabla \boldsymbol{u}(x)) = \mathbf{0}$. Combined with $\operatorname{skw}(\nabla \boldsymbol{u}(x)) = \operatorname{skw} \mathcal{G}_0(x) = \mathbf{0}$, we obtain $\nabla \boldsymbol{u}(x) = \operatorname{sym}(\nabla \boldsymbol{u}(x)) + \operatorname{skw}(\nabla \boldsymbol{u}(x)) = \mathbf{0}$, whence $\boldsymbol{\omega}(x) = \nabla \times \boldsymbol{u}(x) = \mathbf{0}$.

(2) ($\Longrightarrow$) Suppose $\mathrm{TZTPostulates}(s_{\boldsymbol{u}}, \lambda, K)$ holds with $\mathrm{CriticalRegion}(s_{\boldsymbol{u}}, \Lambda) \subseteq K$. If $|\boldsymbol{\omega}(x)|^2 > \Lambda^2$ for some $x \in \mathbb{T}^3$, then $x \in \mathrm{CriticalRegion}(s_{\boldsymbol{u}}, \Lambda) \subseteq K$, so Part (1) implies $\boldsymbol{\omega}(x) = \mathbf{0}$, yielding the contradiction $0 = |\boldsymbol{\omega}(x)|^2 > \Lambda^2 \geq 0$. Hence $|\boldsymbol{\omega}(x)|^2 \leq \Lambda^2$ for all $x \in \mathbb{T}^3$. ($\Longleftarrow$) Conversely, if $|\boldsymbol{\omega}(x)|^2 \leq \Lambda^2$ for all $x \in \mathbb{T}^3$, then $\mathrm{CriticalRegion}(s_{\boldsymbol{u}}, \Lambda) = \emptyset$, so the core postulates on $K = \emptyset$ hold vacuously with $\lambda \equiv 0$.

(3) Since $\|\boldsymbol{\omega}(t)\|_{L^\infty} \leq \Lambda$ on $[0, T]$ and $T \geq 0$, monotonicity of the integral and supremum immediately yields Inequality (24).

(4) For $\boldsymbol{u}^*(x) = (\sin x_2, 0, 0)^{\mathsf{T}}$, $\nabla \cdot \boldsymbol{u}^*(x) = \partial_1(\sin x_2) + 0 + 0 = 0$, and $\boldsymbol{\omega}^*(x) = (0, 0, -\cos x_2)^{\mathsf{T}}$, so $|\boldsymbol{\omega}^*(x)|^2 = \cos^2 x_2 \leq 1 \leq \Lambda^2$ for all $\Lambda \geq 1$ while $|\boldsymbol{\omega}^*(0)|^2 = \cos^2(0) = 1 > 0$. □

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6 Verification Mapping to Lean 4 (Supplementary_2) and SymPy CAS (Supplementary_3)

Table 1 provides the complete cross-reference matrix linking the analytical proofs in this supplementary document, the sections and theorems of the main manuscript (Part1_Paper_v1.5-100.tex), the mechanized Lean 4 formal verification modules (Supplementary_2_Formal_Verification_Lean4), and the SymPy CAS symbolic verification scripts (Supplementary_3_CAS_Python_Proofs).

Table 1: Complete cross-reference matrix linking analytical derivations in Supplementary_1, main text sections/theorems (Part1_Paper_v1.5-100.tex), Lean 4 formal proofs (Supplementary_2), and SymPy CAS symbolic scripts (Supplementary_3).

Supplementary 1	**Main Text Ref.**	**Lean 4 File (Supplementary_2)**	**SymPy CAS File (Supplementary_3)**
Lemma 1.1, Prop. 1.1	Sec. 2.3–2.5, Eqs. (4)–(8), (16)–(17), App. A	—	—
Proposition 2.1	Sec. 3.2, Sec. 4.1, Eqs. (10)–(15)	Theorem0_Basic_Topology.lean	—
Theorem 2.1	Theorem 3.1, Sec. 3.2	Theorem1_Helmholtz_Bridge.lean	CAS_Proof1_Helmholtz_Bridge.py
Theorem 3.1	Sec. 4.2, Eqs. (18)–(21)	Theorem2_Integral_Alignment.lean	CAS_Proof2_Integral_Alignment.py, CAS_Proof4_Viscous_Absorption.py
Prop. 4.2	Remark 3.2, Sec. 4.2	Theorem3_Mean_Conserved.lean	CAS_Proof3_Mean_Conserved.py
Prop. 4.1, Theorem 4.1	Sec. 4.2, Eq. (22)	Theorem4_Viscous_Absorption.lean	CAS_Proof4_Viscous_Absorption.py
Theorem 5.1(4)	Sec. 5.2, Table 2 (Theorem 5)	Theorem5_Consistency_Audit.lean	—
Theorem 5.1(1)–(3)	Eq. (1), Sec. 5.2, Table 2 (Theorem 6)	Theorem6_Vorticity_Ceiling.lean	CAS_Proof1_Helmholtz_Bridge.py

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Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

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